





Early–late asymmetry of the muon density with the UMD

Selection bias, core reconstruction, and how to recover it

Lautaro Silva Pizzi

with J. M. Figueira and F. Sánchez

ITeDA (CNEA-CONICET-UNSAM) — FCEN-UBA

UMD Foundations Meeting — 4 November 2026

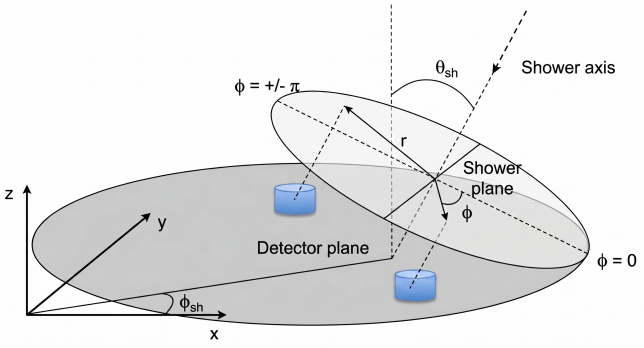
This talk is about the azimuthal asymmetry of the muon density, measured underground with the UMD. I'll show that the asymmetry is clearly there in simulations, then two analysis effects that can hide it or even flip its sign, and finally how we can account for one of them.

Transition: Let me first say what we measure.

The observable: first harmonic in the shower plane

- ϕ: station azimuth in the **shower plane**.
- Vertical shower: circular symmetry.
- Inclined shower: two distinct regions.
  - **early** ($\phi \approx 0°$): reached first, through *less* atmosphere.
  - **late** ($\phi \approx \pm 180°$): reached later, through *more* atmosphere.
- $A_1 > 0$: early excess; $A_1 < 0$: late excess.
- Fitted on an **ensemble** of showers, per (r, θ) bin.

$$\rho(r, \phi) = \langle \rho(r) \rangle \, (1 + A_1 \cos \phi)$$



The asymmetry of the total SD signal in inclined showers is well known. The question here is what happens with muons alone, which is what the UMD gives us: shielded, so free of the electromagnetic component. The observable is the first-harmonic amplitude A1 of the muon density in the shower plane. For a vertical shower the density is circularly symmetric. When the shower is inclined, two regions appear: the early side, reached first through less atmosphere, and the late side, reached later through more atmosphere. A positive A1 means more muons on the early side. A1 is fitted on an ensemble of showers, not event by event.

Transition: We start in the cleanest setup we have.

Why an early–late asymmetry?

Four mechanisms modulate the muon density in the shower plane:

Mechanism	Effect	Favours
Atmospheric attenuation	The late region is farther: more muons decay on the way	**early**
Detector response	Late muons arrive more inclined: a flat detector (UMD) sees a lower density there	**early**
Kinematic divergence	At the same r, the late region is farther ($1/d^2$ dilution) but needs a smaller opening angle (smaller, more probable p_t)	**early** / **late**
Geomagnetic field	Separates μ^+ and μ^-; second order for $\theta \lesssim 70^\circ$	—

A toy model combining them predicts $A_1 > 0$, growing with r and θ.

Why should there be an asymmetry at all? Four mechanisms act on the muons. The late region of an inclined shower is farther from where the muons are produced, so more of them decay on the way: attenuation always favours the early side. The late muons also arrive more inclined, so a flat detector like the UMD sees a lower density there; for the SD tank signal this effect is essentially absent. The third is the kinematic divergence of the muon emission: at the same distance in the shower plane the late stations are farther, which dilutes the density, but they need a smaller opening angle, that is a smaller transverse momentum, which is more probable; the first favours early and the second late. Finally the geomagnetic field separates positive and negative muons, but it is a second-order effect below about seventy degrees and we do not include it. A toy model with a point source of muons on the axis combines the first three and predicts a positive A1, growing with distance and with zenith angle, for both the UMD and the SD. That is the expectation we test. (If asked: to first order in r over D, A1 is about the bracket two minus s plus D over lambda plus kappa, times r over D times tan theta; kappa is one for the UMD plate and zero for the tank signal.)

Transition: Now the simulations.

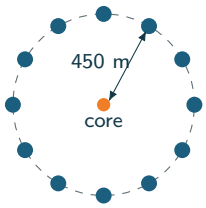
The reference: Monte Carlo geometry

Simulations and the Dense Ring

- Official CORSIKA + *Offline* production, with SD-750 and UMD:

Hadronic models	SIBYLL 2.3e (main), QGSJET-III.01
Primaries	p (main), He, Fe
Energy	$10^{17.5}$–10^{18} eV
Zenith angle	$\theta \leq 65^\circ$

- **Dense Ring**: 12 virtual stations at $r = 450$ m from the **true** axis, every 30° in the shower plane. Same r for all: the radial fall-off of the density cannot produce a modulation in ϕ.

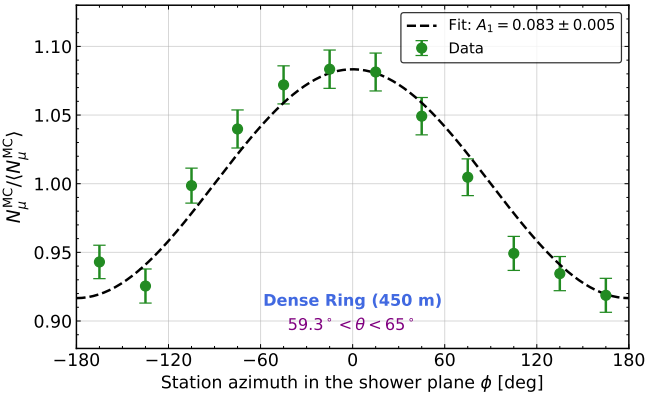


Shower plane.

These are the simulations: the official collaboration production, CORSIKA plus Offline, with the 750 meter array and the UMD. Most of what I show is protons with SIBYLL 2.3e between ten to the 17.5 and ten to the 18 electronvolts. To isolate the effect we start with a synthetic setup, the Dense Ring: twelve virtual stations placed in the simulation at 450 meters from the true shower axis, every thirty degrees. Since all stations are at the same distance, the steep fall of the density with distance cannot produce a modulation in azimuth.

Transition: This is what the muon density looks like on the ring.

A clear modulation on the Dense Ring

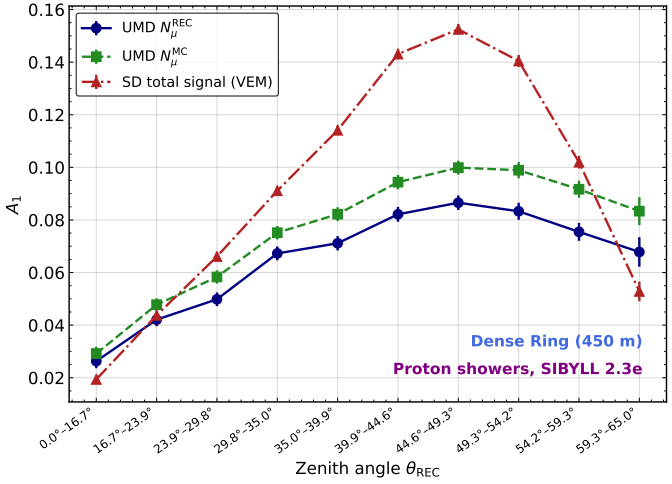


- Injected muon number N_μ^{MC}, $59.3^\circ < \theta < 65^\circ$.
- Maximum in the **early** region ($\phi \approx 0$).
- Early–late difference $\approx 17\%$ peak to peak ($\pm 8\%$ around the mean).

Here are the twelve ring stations for the most inclined bin, using the injected muon numbers. The density peaks on the early side and the first harmonic gives an amplitude of 0.083: the early side has about eight percent more muons than the mean, the late side eight percent fewer. With the reconstructed muon numbers the same bin gives 0.068; that difference is the counting reconstruction, in the backup. (If asked: the Dense Ring zenith bins are in reconstructed zenith angle.)

Transition: How does it change with zenith angle?

Comparison: SD vs. UMD on the Dense Ring



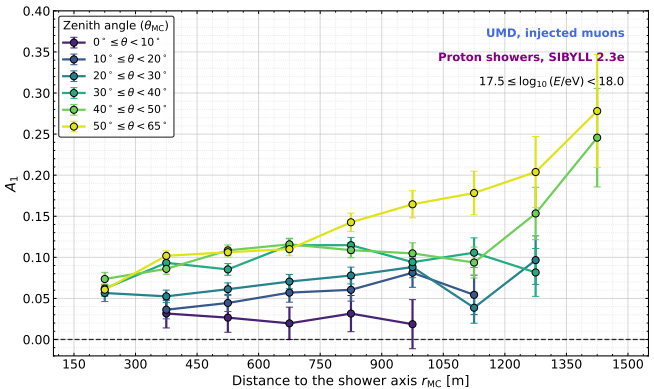
- UMD: $A_1 > 0$, grows with θ up to $\sim$ 45–50°.
- UMD: lower amplitude with reconstructed (N_μ^{REC}) than with injected (N_μ^{MC}) muons, same shape.
- SD total signal: larger than the UMD at $\sim$ 20–60° (EM-dominated); drops sharply above $\sim$ 50°.
- SD below the UMD only at $\theta \gtrsim 60°$.

The muon asymmetry is positive and grows with inclination up to about forty-five to fifty degrees. The SD total signal is larger over most of the range, because it is dominated by the electromagnetic component, which is strongly attenuated after the shower maximum. Above about fifty degrees the electromagnetic part is mostly absorbed and the SD asymmetry drops quickly; only in the last bin, above about sixty degrees, does it fall below the UMD. Note also the gap between the two UMD curves: the reconstructed muon numbers give a lower amplitude than the injected ones, with the same shape. That comes from the muon-counting reconstruction; it is in the backup.

Transition: The ring gives us one radius. For the radial dependence we need the real Infill array.

UMD radial dependence on the Infill (MC geometry)

Real Infill stations (750 m): the distance to the axis varies from station to station $\Rightarrow A_1(r)$. Station r and ϕ depend on where the axis is: here the **true** core and direction.



- **UMD only** (injected muons).
- Radial bins of $\Delta r = 150$ m; zenith bins of 10°.
- $\theta < 10^\circ$: compatible with zero.
- 10–40°: grows with r, then **flattens**.
- $\theta \geq 40^\circ$: **grows with** r.

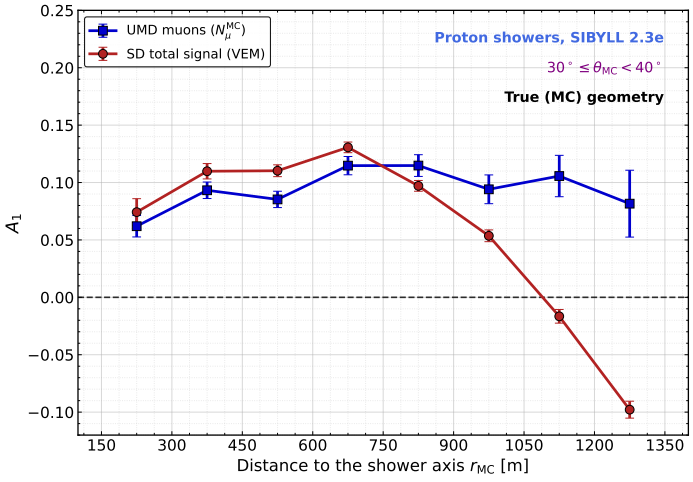
This is our reference.

Still simulations, but now with the real Infill stations. Their distances to the axis vary, so we can follow A1 as a function of r. The distance and azimuth of each station depend on where the shower axis is: we can use the true core and direction from the simulation, or the reconstructed ones, and the main difference between the two is the core position. Here we use the true one. This is the UMD alone, with the injected muon numbers, in radial bins of 150 meters and zenith bins of ten degrees. Near-vertical showers show no significant asymmetry. Between ten and forty degrees it grows with distance and then flattens. Above forty degrees it grows with distance. (If asked: at forty to fifty degrees it stays around 0.1 and rises only beyond about twelve hundred meters, where the statistics are lower.) This is what an ideal analysis should find.

Transition: Now we compare with the surface detector in the same showers.

The SD inversion and the selection bias

The SD comparison: inversion far from the core

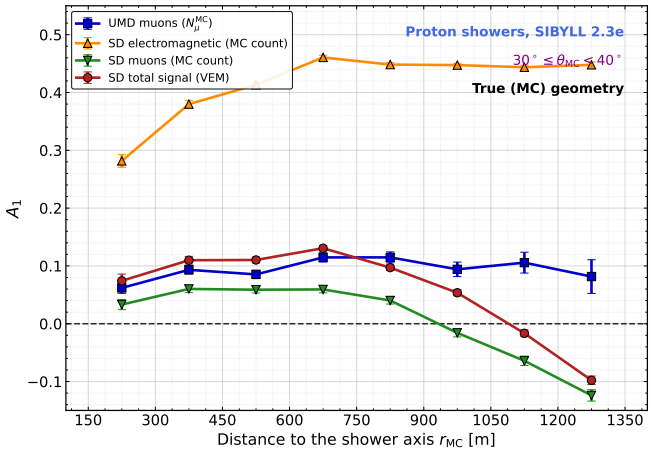


- We now compare the UMD and SD asymmetries in the same Infill showers.
- MC geometry, $30^\circ \leq \theta < 40^\circ$.
- UMD: positive at all r.
- SD total: a maximum, then negative far from the core.

Now we compare the UMD with the surface detector, on the same Infill showers and with the same true geometry. The UMD stays positive at all distances. The total SD signal peaks, then decreases, and far from the core becomes clearly negative, an excess on the late side. Since the core and direction are the true ones, this cannot be a reconstruction error.

Transition: Which part of the tank signal is responsible?

It is the SD muon component that inverts



- MC particle counts in the tank, split by component.
- **EM**: positive at all r.
- **SD muons**: positive near the core, negative beyond $\sim$ 900 m.
- The SD total follows its muons at large r.

Same quantity, opposite sign in UMD and SD.

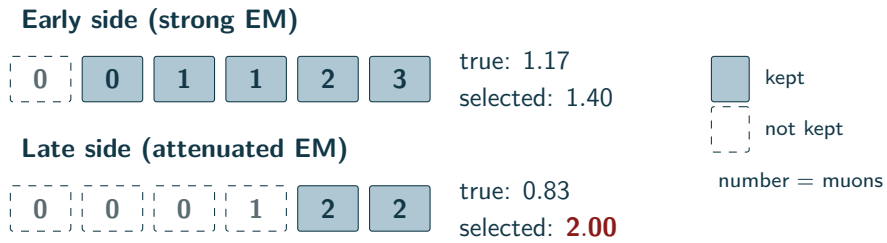
Splitting the tank signal with the simulation truth, the electromagnetic part is positive everywhere. The muon part is what inverts, and the total SD signal follows it far from the core. So we compare the same physical quantity, muons crossing a detector, and get opposite signs in the UMD and in the SD. Our analytic model of muon propagation does not produce a negative asymmetry at these distances, so we looked at how the observable is built.

Transition: The key point: we never average over all stations.

Hypothesis: the station selection biases the average

HYPOTHESIS

A station enters the analysis only if it **belongs to the reconstructed event** (`HasStation`). If the probability that a station is kept depends on its own muon content, and this dependence is not the same on the early and late sides, the **regional averages are biased**.



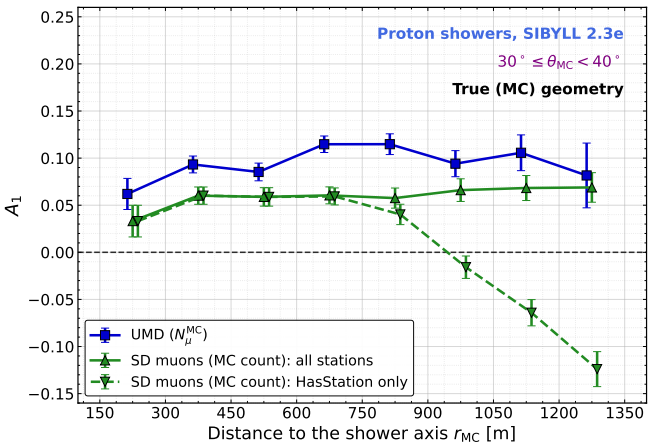
Both averages go up, the late one much more: the sign flips **without moving a single muon**.

Illustrative cartoon. In the simulation, at 900–1500 m, 54% of the early and 33% of the late tanks are kept.

Every curve so far is an average over stations that are part of the reconstructed event. For the UMD that seemed natural, since it has no trigger of its own. But suppose the chance that a station is kept depends on how many muons it has. On the early side the electromagnetic signal is strong, so almost every station is kept, even one with no muons; only a few muon-poor ones are lost and the average goes up a little. On the late side the electromagnetic signal is weaker, so mostly the stations with more muons survive and the average goes up a lot. In this example the true averages are 1.17 and 0.83, an early excess; the selected ones are 1.40 and 2.00, a late excess. The sign flips although no muon was created or moved. This is a hypothesis; the next two slides test it with the simulation.

Transition: The direct test is to remove the requirement.

Test 1: without HasStation, the inversion disappears



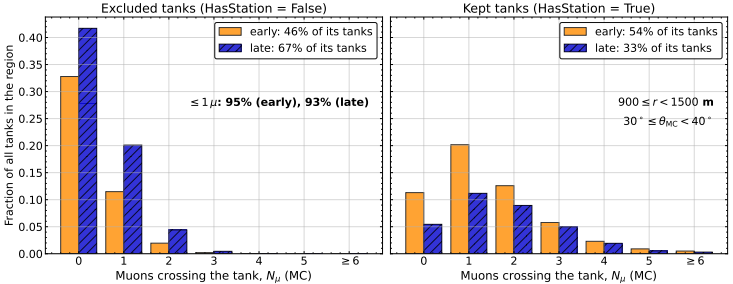
- Same MC counts, coordinates and fit: only **which stations** enter changes.
- All stations: SD muons **positive at all** r ($+0.03$ to $+0.07$).
- Same sign and trend as the UMD, **but not the same values**.

Same Monte Carlo muon counts in the tanks, same coordinates, same fit; only which stations enter changes. Over all simulated stations the SD muon asymmetry is positive at all distances, between plus three and plus seven percent. Requiring the station to be in the reconstructed event changes nothing up to about 750 meters, where practically every station is kept, and then drives the asymmetry down to minus 0.12 at twelve to thirteen hundred meters: plus 0.069 plus or minus 0.016 over all stations, against minus 0.124 plus or minus 0.019 with the requirement. The errors come from a bootstrap over the 960 showers, keeping the stations of each shower together. So the sign flip is a property of the station sample. Without the selection the SD muons have the same sign and trend as the UMD, but between 150 and 900 meters the UMD is still 0.03 to 0.06 higher, and that difference excludes zero at 95 percent (backup slide); detector response and threshold are candidates, this is open.

Transition: If this is a trigger effect, there is a direct check.

Test 2: the excluded tanks should hold very few muons

- If the inversion comes from the **trigger**, the tanks dropped by the reconstruction (`HasStation` = False) should hold, on average, at most one muon: too few for the station to trigger.



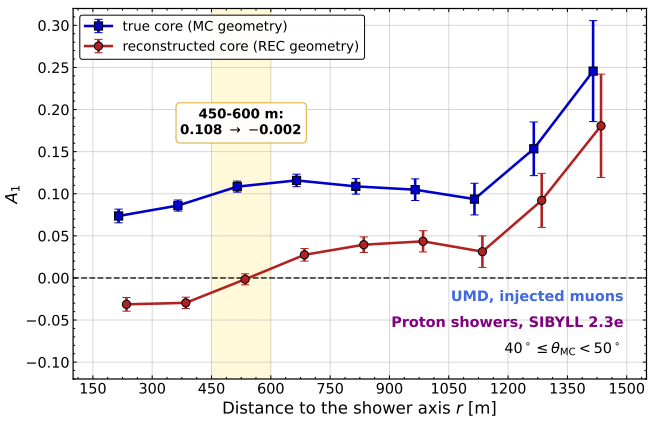
	$\langle N_\mu \rangle$ all tanks	$\langle N_\mu \rangle$ kept tanks	fraction kept
early	0.96	1.49	0.54
late	0.87	1.69	0.33

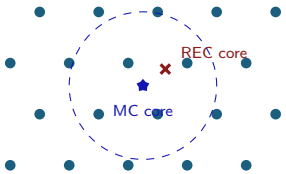
If the inversion is a trigger effect, there is a clear prediction: the tanks that the reconstruction drops should have been crossed by at most about one muon. If we found excluded tanks with several muons, a trigger explanation would not be credible. So we count the Monte Carlo muons in the tanks between 900 and 1500 meters. Each region is normalised to all of its tanks, so the area of the bars is the fraction lost or kept: on the early side 46 percent of the tanks are dropped, on the late side 67 percent. Ninety-five percent of the excluded early tanks and ninety-three percent of the late ones have at most one muon. The table is the bias in numbers: over all tanks the early side has more muons, 0.96 against 0.87, which is the true positive asymmetry; over the kept tanks it is the other way round, 1.49 against 1.69. (If asked why the kept late tanks have more muons than the kept early ones: the selection raises both means, but much more on the late side, by a factor 1.9 against 1.55, because there the electromagnetic signal is weaker, about half, 14 against 28 particles in the excluded tanks, so only muon-rich tanks pass; on the early side tanks with zero or one muon are often kept thanks to their electromagnetic signal. At 1200 to 1350 meters only 18 percent of the late stations are kept, against 42 percent of the early ones.) This supports the trigger picture, but HasStation also includes readout and reconstruction, and we have not separated them, so it is consistent with a trigger effect, not a proof. (Early: $|\phi| < 60°$, late: $|\phi| > 120°$; MC geometry.)

Transition: So far, true core and direction. In real data we only have the reconstructed ones.

Reconstructed geometry and the core position

REC geometry suppresses the asymmetry near the core



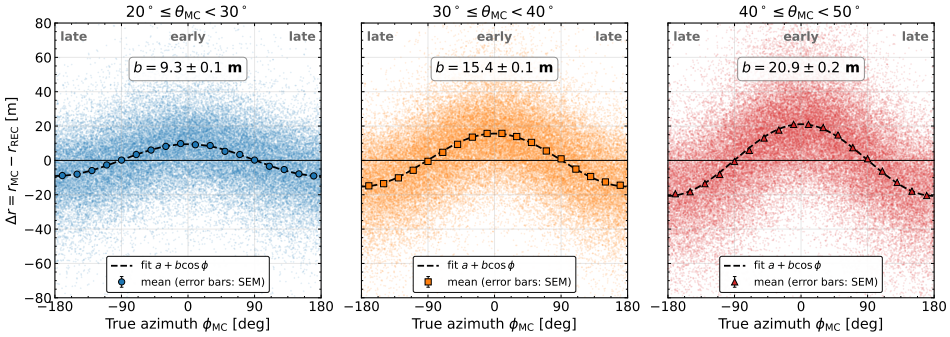


- Same UMD modules and **injected** muons; **only the core changes**.
- Near the core A_1 drops to **zero or below**; farther out the curves **converge**.
- The effect comes from the **offset between the true and the reconstructed core positions**.

This is the reference curve of slide eight, forty to fifty degrees, with the same UMD modules and the same injected muon numbers, now computed a second time with the reconstructed core. Near the core the asymmetry drops to zero or becomes negative: at 450 to 600 meters it goes from 0.108 to minus 0.002. Farther out the two curves approach each other: the loss is not global, it shrinks with distance. We also swapped the two quantities one at a time: using the reconstructed or the true azimuth gives the same A1 within 0.002, so everything comes from the distance to the reconstructed core. (If asked, other zenith bands at 450 to 600 meters: 0.061 to 0.011 at twenty to thirty degrees, 0.085 to 0.002 at thirty to forty.)

Transition: So where is the reconstructed core?

The reconstructed core is shifted toward the early region



- $\Delta r = r_{\mathrm{MC}} - r_{\mathrm{REC}}$ vs. ϕ is fitted by $a + b\cos\phi$, the modulation expected from a **rigid displacement of the core** by b: early stations are placed **closer** to the core, late ones **farther**; $a \simeq 0.2$ m.
- Likely cause: the reconstruction LDF is **symmetric in** ϕ; it absorbs the early excess by **moving the core toward the early side, where there is more signal**.

For each station we compare its distance to the true and to the reconstructed core, as a function of its true azimuth. If the core error were random, this would be flat. Instead it follows a cosine: early stations are placed closer to the core than they are, late stations farther. That is the signature of a systematic shift of the reconstructed core toward the early side: if the core is displaced rigidly by b toward the early direction, to first order every station's distance changes by b cos phi, which is exactly the fitted form. The constant a comes out at about 0.2 meters, so there is no azimuth-independent radial bias. The error bars are the standard error of the mean in each azimuth bin, smaller than the markers; the cloud behind them is the station-by-station spread, about twenty meters. So the modulation is very well determined even though single stations scatter a lot. The fitted shift is 9.3, 15.4 and 20.9 meters, with errors of 0.1 to 0.2 meters. (If asked: those errors treat the stations as independent, but the stations of one shower share the same core error, so they are underestimated; a bootstrap over events is pending. The trend with zenith angle is far larger than any plausible error.) A likely reason: the LDF used in the reconstruction is symmetric in azimuth. Faced with more signal on the early side, the only thing the fit can do is move the core toward it. (Infill stations with reconstructed distance below 1400 meters.)

Transition: What does a shift of this size do to the LDF, and to A1?

A shifted core adds a $\cos\phi$ term to the LDF

- Binned by r_{REC}, a station at r is really at $r + b\cos\phi$: its density belongs to **another radius**.

$$\rho(r + b\cos\phi) \simeq \rho(r)\Big[1 - \beta\,\frac{b}{r}\cos\phi\Big] \;\Rightarrow\; \boxed{\Delta A_1 \simeq -\beta\,\frac{b}{r}} \qquad \beta = -\frac{\mathrm{d}\ln\rho}{\mathrm{d}\ln r}\ \text{(local)}$$

$450 \le r < 600$ m, injected muons

θ	b [m]	β	ΔA_1 measured	$-\beta b/r$ predicted	residual
20–30°	9.3	2.96	−0.050	−0.052	+0.002
30–40°	15.4	2.72	−0.084	−0.080	−0.004
40–50°	20.9	2.52	−0.110	−0.100	−0.010

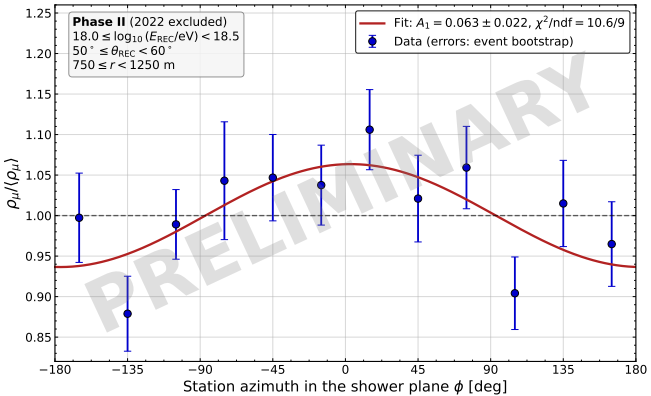
Here is why a shifted core changes the asymmetry. We bin stations by their reconstructed distance. From the previous slide, a station that we place at distance r is really at r plus b cos phi: an early station is farther than we think, a late one closer. So its density is the density of another radius, and that difference has a cos phi dependence. Locally the LDF falls with slope beta, minus the logarithmic derivative of the density at the radius of the bin; it is not a global exponent. To first order the density gets a factor one minus beta b over r cos phi: a spurious cosine term of opposite sign to the physical one, which the harmonic fit absorbs into A1. So the measured A1 should drop by beta b over r. The table, all at 450 to 600 meters with injected muons, puts measured and predicted side by side: b comes from the previous slide, beta is measured in the same bin, and r is about 526 meters, the harmonic mean of the reconstructed distances. Both b and beta are measured on their own, before looking at the asymmetry, so the prediction has no free parameter tuned to match the loss; the residuals are below 0.01, about ten percent of the loss. (Sign, if asked: Delta r is r MC minus r REC equal to b cos phi, so r MC equals r REC plus b cos phi; the argument of rho is the true distance of a station binned at r REC, so the plus sign is correct.)

Transition: So what would it take to apply this to real data?

A first look at field data

The modulation is visible in field data (preliminary)

FIELD DATA – PRELIMINARY



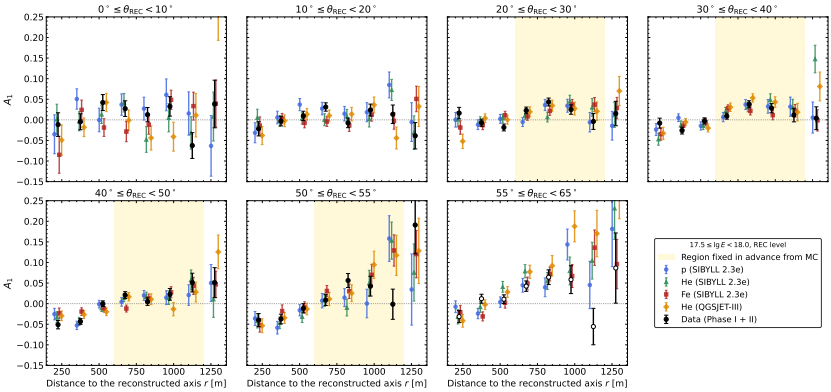
- UMD muon density vs. ϕ, normalised to its mean, Phase II; REC geometry.
- Maximum at $\phi \approx 0$: **early excess**, as in the simulations.
- $A_1 = 0.063 \pm 0.022$ (2.8σ), $\chi^2/\mathrm{ndf} = 10.6/9$; errors from a bootstrap over **events**.
- **Preliminary**: no core-shift correction, no systematics, no MC above 10^{18} eV.

This is the UMD muon density measured in the field, as a function of azimuth in the shower plane, for Phase II events in the cell shown in the box. The maximum is on the early side, as in the simulations. The first harmonic gives 0.063 plus or minus 0.022. That error comes from a bootstrap over events, because the modules of one shower are correlated; treating every module as independent gives 0.017, which overstates the significance. So this is a 2.8 sigma early excess, with a good fit. It is preliminary: the core correction we just discussed is not applied, there is no systematics study, and our simulations stop at ten to the eighteen. (If asked: part of the zenith range, 55 to 60 degrees, is outside the standard SD-750 energy calibration; and 2022 Phase II commissioning data are excluded.)

Transition: And the radial pattern?

Field data follow the REC-geometry simulations

FIELD DATA – PRELIMINARY



- Data (Phase I + II) and simulations with the **same reconstruction**, $17.5 \leq \lg E < 18.0$.
- Same radial pattern: $A_1 \leq 0$ near the core in inclined showers, positive farther out: the **signature of the core shift**, seen in data.

Here are all the cells, data in black and simulations processed with the same reconstruction in colours, one panel per zenith band. The data follow the simulations, including the feature we have just explained: in inclined showers the asymmetry is negative or zero near the core and becomes positive farther out. That is the prediction of the core shift, beta b over r, which falls with distance, and here it appears in data. The yellow band is the region where we expected the asymmetry to survive the reconstruction, fixed from the simulations before looking at the data. This is not a composition measurement: at this statistics and with REC geometry the primaries are not separated.

Transition: Let me summarize.

Conclusions

Conclusions

1. In simulations the UMD muon density has a **positive early–late asymmetry**, growing with r for inclined showers.
2. The SD muon asymmetry inverts far from the core **only** when averaging over the stations kept by the reconstruction (a **station-by-station** selection); over all stations it is positive. Excluded tanks hold **mostly** ≤ 1 muon: **consistent with** a trigger effect.
3. With the reconstructed geometry the asymmetry is **suppressed near the core**: the reconstructed core is shifted toward the early region by $b \approx 9$–21 m.
4. The loss follows $\Delta A_1 \simeq -\beta\, b/r$; adding it back recovers the MC reference. Applying it to data requires b **without MC truth**.
5. Field data (preliminary): early excess, $A_1 = 0.063 \pm 0.022$, and a radial pattern that follows the REC-geometry simulations.

Data–MC comparisons need the same station selection and the same geometry.

Next: parametrize $b(\theta)$, $\beta(\theta)$ and validate them on independent simulations; correct the full radial range and field data; separate trigger, readout and reconstruction in `HasStation`.

Five points. The muon asymmetry is there, and it behaves as expected with zenith angle and distance. What we measure depends on how we build the observable: which stations we average over can flip the sign, and the reconstructed core can erase it. The second effect is simple and predictable, so it can be accounted for, and any comparison between data and simulation must use the same station selection and geometry. The open step is to determine the core shift without Monte Carlo truth. Next, we want to parametrize b and beta as functions of zenith angle, at this fixed energy, and check on simulations not used to build it that the correction recovers the asymmetry; apply it over the whole radial range and to field data; and separate trigger, readout and reconstruction inside HasStation. (If asked how to get b in data: either from that parametrization, validated on independent simulations of other primaries and hadronic models, which tests the model and composition dependence but not data-MC differences such as the muon deficit; or by removing the shift at its origin, with a reconstruction LDF that allows an azimuthal modulation, which is hard event by event at Infill energies but worth trying for the most energetic events. The closure test so far is one radial bin, protons, one hadronic model, injected muons; the UMD counting bias is a separate, smaller effect.)

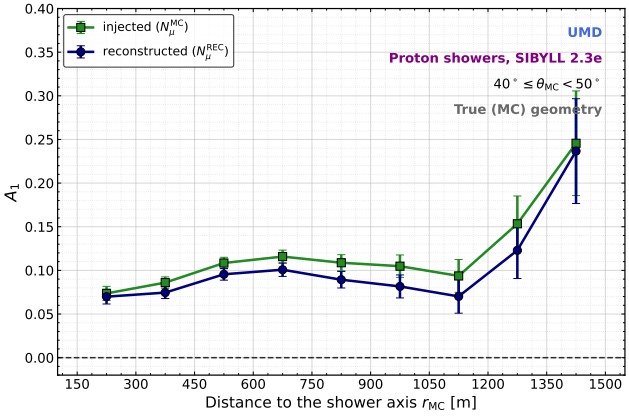


Thank you!

Questions?

Image: A. Chantelauze, S. Staffi, L. Bret (Pierre Auger Observatory)

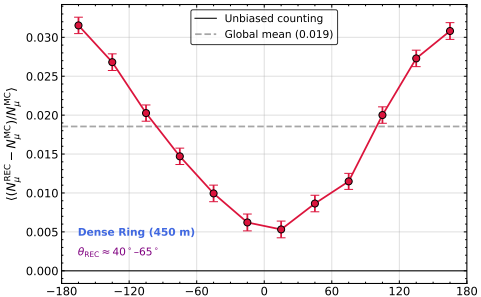
Backup: muon-counting reconstruction (MC geometry)

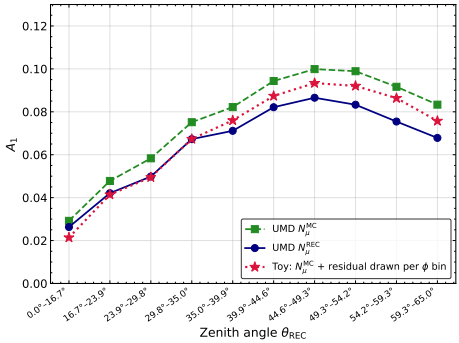


- 40°–50°, true core and direction.
- The reconstructed muon number lowers A_1 by 5–23%, with the same shape.
- A separate effect from the core shift.

Backup: the counting bias depends on azimuth

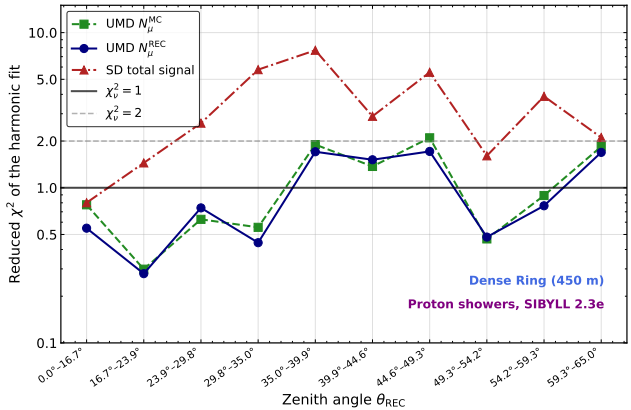
SIMULATIONS – DENSE RING





- Left: mean relative residual $(N^{REC} - N^{MC})/N^{MC}$ vs. ϕ: ≈ 0.005 early, ≈ 0.03 late. More over-counting where there are fewer muons $\Rightarrow$ lower A_1.
- Right: injecting the residuals drawn per ϕ bin into N_μ^{MC} reproduces A_1^{REC} up to $\theta \approx 35^\circ$; above, half to two thirds of the loss. The remnant is open.

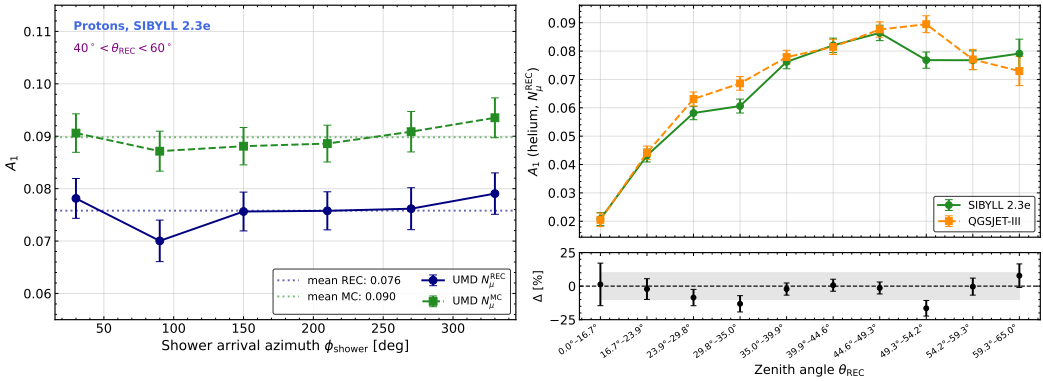
Backup: fit quality on the Dense Ring



- UMD, MC and REC: $\chi^2_\nu \approx 1$ (0.3 to 2.1).
- SD: χ^2_ν up to ~ 8. With such small errors the fit is sensitive to higher harmonics and to correlations between stations of one shower.

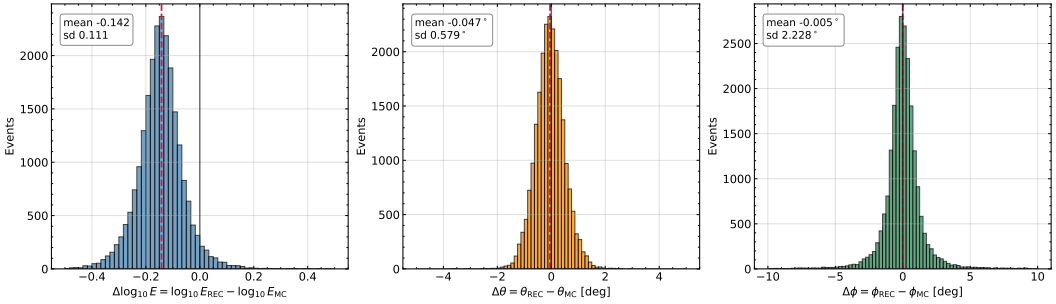
Backup: robustness

SIMULATIONS – DENSE RING



- Left: A_1 does not depend on the arrival azimuth of the shower.
- Right: helium, SIBYLL 2.3e vs. QGSJET-III: within $\pm 10\%$ over most of the range.

Backup: reconstruction performance



- Energy: mean -0.142, spread 0.111 in $\Delta \log_{10} E$ (fewer muons in the simulation than in data, with a data-based energy calibration).
- θ: bias $-0.05°$, spread $0.58°$. ϕ: bias $-0.005°$, spread $2.2°$.

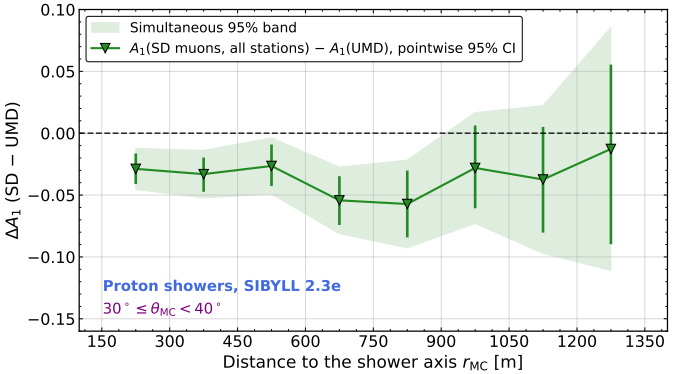
Backup: the core shift and the Dense Ring

- In reconstructed coordinates the ring **is** eccentric, but the stations are physically at 450 m from the **true** axis, and they are not re-binned in r.
- The assigned azimuth rotates by $\sim b/r \approx 2^\circ$, much less than the 30° spacing; the assigned radius changes by at most $b \approx 20$ m.
- Both affect N_μ^{MC} and N_μ^{REC} in the same way, so they cannot explain the MC–REC gap on the ring (that gap is the counting bias).

Backup: is the UMD biased by the selection too?

- The UMD has no trigger of its own: a module is read only if the tank of its station is kept. So the UMD curve is also averaged over `HasStation` stations.
- But the selection acts on the **tank's own particles** (its muons and its EM signal). The module counts **other muons**: a different area, not right under the tank, behind $\sim$2.3 m of soil ($\sim$1 GeV threshold).
- Given the station position, the expected UMD count is still the local muon density: to first order, **no bias**.
- A residual effect can enter through what tank and module share, the fluctuation of the local density from shower to shower and station to station. Not quantified here; the UMD count of the stations that were not kept is not stored in this production.

Backup: SD muons over all stations vs. UMD



- Without `HasStation`, the SD muons stay **below** the UMD: $\Delta A_1 \approx -0.03$ to -0.06 between 150 and 900 m.
- There the pointwise 95% interval excludes zero; beyond 900 m the difference is not resolved.
- Paired bootstrap over the 960 showers (UMD and SD from the same showers).
- Candidates: flat plate vs. tank response, UMD threshold. Open.